CHAPTER: 1 — The Skin and Systemic Disease

Professional Clinical Study Guide

OWL Medical Education

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First Edition

Chapter 1

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SUBTITLE: Cutaneous Manifestations of Internal Disease — A Comprehensive Clinical Review

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SECTION: Overview: Clues to Systemic Disease on Skin Examination

The skin is a mirror of internal health. Recognizing patterns of cutaneous disease is a critical clinical skill for identifying underlying systemic conditions. When a rash is refractory to topical treatment or associated with constitutional symptoms, clinicians must broaden their diagnostic lens. ---

Chapter 2

SUBSECTION: Red Flags — Features Suggesting Systemic Disease

SUBSECTION: Red Flags — Features Suggesting Systemic Disease

Not every rash is dermatology. The following clinical clues should prompt evaluation for systemic disease:

KEY TEACHING POINTS

- The mnemonic key**: rash **ass**ociated with joint pain, fever, weight loss, weakness, SOB, altered bowel function → Think systemic vasculitis, autoimmune disease, or malignancy.
- Non-blanching palpable purplish fixed lesions that are painful/blistering** → Classic for **vasculitis** — indicates inflammatory destruction of vessel walls, not a simple allergic rash.
- Rash not responding to topical treatment** → Always consider secondary causes: internal malignancy, IBD, sarcoidosis, infection, or endocrine disorders.
- Erythema with dermal palpable lesions** → Discrete dermal nodules suggest granuloma, lymphoma, or metastatic disease — biopsy is essential.
- Unusual pigmentation or texture changes** → Acanthosis nigricans, ichthyosis, hyperpigmentation → Screen for malignancy, insulin resistance, Addison's disease.
- > ■ **Test Alert**: Non-blanching palpable purpura = **True Vasculitis until proven otherwise**

MCQ TRAPS & PITFALLS

- Do NOT confuse blanching maculopapular rash (toxic erythema) with non-blanching palpable purpura (vasculitis)** — the diascopy test is the key differentiator.

• A rash that fails topical therapy is NOT treatment-resistant dermatitis** — always investigate for systemic causes before escalating topical potency.

• Dermal nodules are NOT always benign cysts** — consider granuloma, lymphoma, and metastatic disease in the differential.

Chapter 3

SECTION: Characteristic Rashes That Indicate Underlying Systemic Disease

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TABLE: Pathognomonic Skin Lesions and Their Systemic Associations

Skin Lesion	Morphology	Key Systemic Associations	Most Common Trigger
Erythema multiforme	Target lesions (concentric erythema, 3 zones of color)	HSV, Mycoplasma pneumoniae, Hepatitis B/C	**HSV (most common overall)**
Pyoderma gangrenosum	Painful enlarging ulcer with undermined violaceous borders	IBD (Crohn's/Ulcerative Colitis), Rheumatoid Arthritis, Hematological malignancy	IBD
Erythema nodosum	Tender subcutaneous nodules (shins)	IBD, Streptococcal infection, TB, Sarcoidosis, Behçet's disease, drugs	**Streptococcal infection**
Vasculitis (palpable purpura)	Non-blanching purpuric papules/plaques	Hepatitis B/C, SLE, Lymphoma, Leukemia, cryoglobulinemia	Depends on context

IMPORTANT

• Erythema nodosum is the **most common panniculitis** — affects subcutaneous fat (septal inflammation without vasculitis).

• Pyoderma gangrenosum is a **diagnosis of exclusion** — must rule out infection, vasculitis, and other causes of ulceration.

• EM is **NOT** a Type I hypersensitivity reaction — it is a **Type IV (delayed) hypersensitivity reaction**.

• Systemic symptoms of the underlying infection (e.g., HSV) usually **precede the EM rash by 2–14 days**.

Chapter 4

OSCE PEARLS

OSCE PEARLS

• When you see **target lesions**, immediately ask about **recent HSV infection or Mycoplasma pneumonia** — these are the two most common triggers.

- Pyoderma gangrenosum** shows **pathergy** (worsening with surgical debridement) — do NOT debride these ulcers.
- Erythema nodosum** is a **reactive process** — always search for the underlying cause, especially in recurrent cases.

QUESTION:

Q: A 28-year-old woman presents with a 3-day history of painful, raised, purplish lesions on her lower legs that do not blanch on pressure. She also reports joint pain, fever, and weight loss over the past 2 weeks. Urinalysis shows hematuria. What is the most likely diagnosis? A) Allergic contact dermatitis B) Erythema nodosum C) Leukocytoclastic vasculitis D) Cellulitis E) Psoriasis ANSWER: C EXPLANATION: The key features pointing to **leukocytoclastic vasculitis** are: (1) **non-blanching palpable purpura** — this indicates vessel wall inflammation and extravasation of RBCs, distinguishing it from simple erythema; (2) systemic symptoms (joint pain, fever, weight loss); (3) end-organ involvement (hematuria suggesting renal vasculitis). Erythema nodosum (B) presents as tender subcutaneous nodules, not purpuric lesions. Allergic contact dermatitis (A) is pruritic and blanching. Cellulitis (D) is typically unilateral, warm, and tender — not purpuric. Psoriasis (E) presents as scaly plaques, not purpura. Vasculitis is associated with Hepatitis B/C, SLE, lymphoma, and leukemia — investigations should include hepatitis serology, ANA, and complement levels. ---

CHAPTER: 2 — Skin Reactions Associated with Infections

Chapter 5

SUBTITLE: Infectious and Inflammatory Exanthems

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SECTION: 1. Toxic Erythema (Exanthematous / Morbilliform Drug Eruption)

A common, self-limiting, widespread eruption triggered by infections or medications.

SUBSECTION: Clinical Features

- Widespread symmetrical maculopapular erythema (morbilliform pattern)
- Begins on the trunk → spreads centrifugally to limbs
- Usually asymptomatic but may be pruritic
- Blanching** on diascopy (key distinguishing feature from purpura/vasculitis)
- Duration: Typically 7–14 days

Chapter 6

SUBSECTION: Etiology

SUBSECTION: Etiology

Viruses (most common), bacteria, and drugs. **Classic presentation**: **EBV (glandular fever) treated with ampicillin → up to 90% develop rash**. This occurs because EBV sensitizes the immune system to penicillin-class antibiotics. Other viral triggers include HHV-6 (roseola), Hepatitis B/C, and HIV seroconversion.

KEY TEACHING POINTS

IMPORTANT

- Toxic erythema is a **Type IV hypersensitivity reaction** — NOT Type I (no anaphylaxis risk).

- The **ampicillin + EBV** association is one of the highest-yield facts in medical exams — up to 90% of EBV patients given ampicillin develop a rash.

- **Diascopy (glass press test)**: Blanching = toxic erythema; Non-blanching = vasculitis. This is the single most important bedside test.

- Management is **supportive** — withdrawal of offending drug, emollients, and antihistamines for pruritus. Topical steroids may help symptomatically.

MCQ TRAPS & PITFALLS

• Do NOT confuse toxic erythema with early Stevens-Johnson Syndrome (SJS)** — SJS has mucosal involvement, skin pain, and systemic toxicity. Toxic erythema is benign and self-limiting.

• A morbilliform rash in a patient on ampicillin does NOT always mean drug allergy** — consider EBV co-infection as the true cause.

• Toxic erythema does NOT cause skin peeling or desquamation** — if present, consider scarlet fever or Kawasaki disease.

Chapter 7

SECTION: 2. Erythema Multiforme (EM)

SECTION: 2. Erythema Multiforme (EM)

An acute, self-limiting, immune-mediated reaction characterized by pathognomonic target lesions.

SUBSECTION: Clinical Features

• Target lesions** (iris lesions): Concentric rings of color change — typically 3 zones (dark center, pale edematous ring, erythematous outer ring)

• Predilection for **acral sites**: Dorsum of hands, feet, elbows, knees

• May have **mucosal involvement** (mild — oral erosions) in EM major

• Self-limiting: Resolves in 2–4 weeks

• May be recurrent (especially with HSV reactivation)

SUBSECTION: Etiology

• HSV (Herpes Simplex Virus)** — most common trigger overall (especially recurrent EM)

• Mycoplasma pneumoniae** — most common trigger in children

• Other infections: Hepatitis B/C, EBV, CMV, Histoplasmosis

• Drugs: Sulfonamides, phenytoin, barbiturates, penicillin

• Malignancy and collagen vascular diseases (rare)

Chapter 8

SUBSECTION: Pathogenesis

SUBSECTION: Pathogenesis

EM is a **Type IV (delayed) hypersensitivity reaction** — immune complex-mediated damage to keratinocytes triggered by infectious antigens or drug metabolites. HSV DNA can be detected in keratinocytes even without active lesions.

KEY TEACHING POINTS

• Target lesions are pathognomonic** — if you see them, think EM until proven otherwise.

• EM minor**: Target lesions only, no mucosal involvement or <10% BSA

• EM major**: Target lesions + mucosal involvement, but <10% BSA, no systemic toxicity

• EM is NOT Stevens-Johnson Syndrome** — SJS involves >10% BSA, severe mucosal erosions, systemic toxicity, and skin detachment. The distinction is critical for prognosis.

• Recurrent EM** → Think HSV → Consider prophylactic acyclovir 500mg PO daily

TABLE: Erythema Multiforme vs. Stevens-Johnson Syndrome vs. Toxic Epidermal Necrolysis

Feature	Erythema Multiforme	SJS	TEN
BSA involvement	<10%	<10%	>30%
Target lesions	Present (pathognomonic)	Atypical target lesions	Absent
Mucosal involvement	Mild (oral only)	Severe (≥2 mucosal sites)	Severe
Skin detachment	None	<10%	>30%
Systemic toxicity	Absent	Present	Severe
Mortality	<1%	5–10%	25–35%
Most common cause	HSV	Drugs (sulfonamides, anticonvulsants)	Drugs
Management	Supportive ± acyclovir	Supportive, consider IVIG/Cyclosporine	Burn unit, supportive

Chapter 9

MCQ TRAPS & PITFALLS

MCQ TRAPS & PITFALLS

MCQ TRAPS & PITFALLS

- **EM is NOT caused by a Type I hypersensitivity reaction** — it is Type IV. This is a classic exam trap.

- **EM with mucosal involvement is still EM major, NOT SJS** — the key differentiator is BSA involvement and systemic toxicity.

- **Do NOT give acyclovir for acute EM** — it does not alter the acute episode. Acyclovir is only useful for **recurrent EM prophylaxis**.

OSCE PEARLS

OSCE PEARLS

- In an OSCE station with target lesions, **always examine the mouth** for mucosal involvement to distinguish EM minor from EM major.

- **Ask about recent cold sores** — HSV is the most common cause of recurrent EM.

- **Do NOT biopsy target lesions** — diagnosis is clinical. Biopsy is only needed if the diagnosis is uncertain.

QUESTION:

Q: A 22-year-old university student presents with a 5-day history of painful, target-shaped lesions on his hands and feet. He reports having a cold sore 10 days prior to the rash. Oral examination shows mild erosions on the buccal mucosa. There is no systemic toxicity and BSA involvement is estimated at 5%. What is the most likely diagnosis? A) Stevens-Johnson Syndrome B) Erythema multiforme major C) Pemphigus vulgaris D) Hand-foot-and-mouth disease E) Secondary syphilis ANSWER: B EXPLANATION: This presentation is classic for **Erythema multiforme major**: (1) **target lesions** on acral sites (hands and feet); (2) **preceding HSV infection** (cold sore 10 days prior — the typical 2–14 day interval); (3) **mild mucosal involvement** (oral erosions); (4) **<10% BSA** and **no systemic toxicity** — this rules out SJS (A). Pemphigus vulgaris (C) presents with flaccid blisters and positive Nikolsky sign, not target lesions. Hand-foot-and-mouth disease (D) typically affects children and causes painful oral ulcers with vesicular lesions on hands/feet — not target lesions. Secondary syphilis (E) causes a generalized copper-colored maculopapular rash, not target lesions. Management is supportive; consider prophylactic acyclovir if episodes recur. ---

SECTION: 3. Erythema Nodosum (EN)

SECTION: 3. Erythema Nodosum (EN)

The most common form of panniculitis — inflammation of subcutaneous fat.

SUBSECTION: Clinical Features

- Tender, erythematous, subcutaneous nodules** — typically bilateral on the **shins** (pretibial area)
- Warm and painful** to palpation
- Overlying skin is erythematous and smooth (no ulceration)
- Septal panniculitis** on histology (inflammation of fibrous septa between fat lobules — **without vasculitis**)
- Associated systemic symptoms: Fever, malaise, arthralgia
- Resolves spontaneously in 3–6 weeks (often with characteristic color changes: red → purple → yellow/green — "bruise-like")
- Does NOT ulcerate** — if ulceration is present, consider other diagnoses

SUBSECTION: Etiology

Chapter 11

TABLE: Causes of Erythema Nodosum

TABLE: Causes of Erythema Nodosum

Category	Specific Causes
Infections (most common)	Streptococcal pharyngitis (#1), TB, Leprosy, Mycoplasma, Yersinia, Histoplasmosis, Coccidioidomycosis
Inflammatory/Autoimmune	IBD (Crohn's > UC), Sarcoidosis (#1 non-infectious cause), Behçet's disease
Drugs	Sulfonamides, Penicillins, Oral contraceptive pill, Bromides
Malignancy	Lymphoma, Leukemia (rare)
Pregnancy	Hormonal influence
Idiopathic	Up to 55% of cases — diagnosis of exclusion

KEY TEACHING POINTS

- Streptococcal infection is the most common cause overall** — always check ASO titer and throat swab.
- Sarcoidosis is the most common non-infectious cause** — check CXR (bilateral hilar lymphadenopathy), serum ACE, calcium.
- IBD association**: EN parallels disease activity in Crohn's disease — may precede GI symptoms.
- EN is a reactive process** — it is NOT an infection of the skin. Treatment is directed at the underlying cause.

• Biopsy is NOT routinely needed** — diagnosis is clinical. Biopsy shows septal panniculitis without vasculitis.

MCQ TRAPS & PITFALLS

• EN does NOT ulcerate** — if you see ulceration, think pyoderma gangrenosum, vasculitis, or infection.

• EN is NOT a vasculitis** — it is a septal panniculitis. This is a common exam confusion point.

• Bilateral shins is the classic location** — unilateral EN should raise concern for localized infection or trauma.

Chapter 12

OSCE PEARLS

OSCE PEARLS

• Always examine the throat and check ASO titer** — streptococcal pharyngitis is the #1 cause.

• Ask about diarrhea and abdominal pain** — EN may be the presenting feature of undiagnosed IBD.

• CXR is essential** — to rule out sarcoidosis and TB.

• Treatment**: NSAIDs (e.g., Ibuprofen 400mg PO TID) for symptom relief, potassium iodide solution for refractory cases, and treat the underlying cause.

QUESTION:

IMPORTANT

Q: A 35-year-old woman presents with a 1-week history of painful, red, swollen nodules on both shins. She reports a sore throat 2 weeks ago. She also has intermittent diarrhea and abdominal pain. ESR is elevated at 65 mm/hr. What is the most appropriate next investigation?

A) Skin biopsy

B) Colonoscopy

C) Throat swab and ASO titer

D) Chest X-ray

E) ANA and anti-dsDNA

ANSWER: C

EXPLANATION: The presentation is classic for **erythema nodosum** — tender bilateral shin nodules with preceding sore throat. The **most common cause of EN is streptococcal infection**, so the **first-line investigation** is throat swab and ASO titer (C). While the diarrhea and abdominal pain raise the possibility of IBD (B), streptococcal infection is statistically more likely and should be investigated first. Skin biopsy (A) is not routinely needed — EN is a clinical diagnosis. CXR (D) is important to rule out sarcoidosis and TB but is not the first step. ANA (E) would be relevant if SLE were suspected, but EN is not a typical feature of SLE. A stepwise approach: (1) Rule out streptococcal infection, (2) If negative, investigate for IBD and sarcoidosis.

SECTION: 4. Pyoderma Gangrenosum (PG)

A rare, neutrophilic dermatosis characterized by rapidly progressive, painful ulceration.

Chapter 13

SUBSECTION: Clinical Features

SUBSECTION: Clinical Features

• Painful, rapidly enlarging ulcer** with characteristic **undermined violaceous (purple) borders**

• Pathergy**: Lesions worsen with surgical debridement or minor trauma (pathognomonic feature)

• Common sites: Lower legs (most common), but can occur anywhere including surgical wounds and stomas

• May start as a pustule or nodule → rapidly progresses to ulcer

• Diagnosis of exclusion** — must rule out infection, vasculitis, malignancy, and other causes of ulceration

• Associated with systemic disease in 50–70% of cases

SUBSECTION: Etiology and Associations

TABLE: Systemic Associations of Pyoderma Gangrenosum

| Association | Frequency | Notes | |---|---|---| | **IBD (Crohn's disease, Ulcerative Colitis)** | Most common systemic association | PG may parallel or precede IBD flares | | **Rheumatoid Arthritis** | Common | Especially seropositive RA | | **Hematological malignancy** | Common | Myelodysplastic syndrome, AML, multiple myeloma | | **Hepatitis (B/C)** | Less common | | | **Sarcoidosis** | Rare | | | **PAPA syndrome** | Rare genetic | Pyogenic Arthritis, PG, Acne — autosomal dominant | | **Idiopathic** | 25–50% | No underlying cause identified |

Chapter 14

KEY TEACHING POINTS

KEY TEACHING POINTS

• Pathergy is pathognomonic** — do NOT debride PG ulcers. Surgical intervention worsens the condition.

• PG is a diagnosis of exclusion** — always biopsy to rule out infection (bacterial, fungal, mycobacterial), vasculitis, and malignancy.

• PG may be the first manifestation of undiagnosed IBD** — always ask about GI symptoms.

• Treatment**: High-dose systemic corticosteroids (Prednisolone 1mg/kg/day), immunosuppressants (Cyclosporine, Mycophenolate), biologics (Infliximab, Adalimumab). Topical Tacrolimus for mild cases.

MCQ TRAPS & PITFALLS

• Do NOT confuse PG with venous ulcers** — venous ulcers have sloping borders, not undermined violaceous borders.

• Do NOT debride PG ulcers** — pathergy will cause worsening. This is a critical management point.

• PG is NOT an infection** — despite the name "pyoderma," it is a neutrophilic dermatosis, not a bacterial infection.

OSCE PEARLS

• Always ask about bowel symptoms** — diarrhea, blood in stool, abdominal pain → think IBD.

• Check for pathergy** — ask if the lesion worsened after any procedure or trauma.

• Biopsy the edge of the ulcer** (not the center) — shows neutrophilic infiltrate without organisms.

• Do NOT start antibiotics** — PG is not infectious. Antibiotics will not help and may delay correct treatment.

Chapter 15

QUESTION:

QUESTION:

Q: A 45-year-old man with known Crohn's disease presents with a rapidly enlarging, painful ulcer on his left lower leg. The ulcer has purple, undermined borders. He reports that it started as a small pustule 5 days ago and has rapidly expanded. He underwent surgical debridement at another hospital, after which the ulcer worsened significantly. What is the most likely diagnosis? A) Venous ulcer B) Cutaneous Crohn's disease C) Pyoderma gangrenosum D) Cellulitis with abscess formation E) Squamous cell carcinoma ANSWER: C EXPLANATION: This is a classic presentation of **pyoderma gangrenosum**: (1) **rapidly enlarging painful ulcer** with **undermined violaceous borders**; (2) **pathergy** — worsening after surgical debridement (pathognomonic); (3) **association with Crohn's disease** (IBD is the most common systemic association); (4) **rapid progression from pustule to ulcer** over 5 days. Venous ulcers (A) have sloping borders, not undermined purple borders, and do not worsen with debridement. Cutaneous Crohn's disease (B) can cause ulceration but does not typically show pathergy. Cellulitis (D) would show warmth, erythema, and systemic signs of infection — not undermined borders. SCC (E) grows slowly and does not show pathergy. Management: **Stop debridement**, start high-dose Prednisolone 1mg/kg/day, and consider Infliximab for refractory cases. ---

SECTION: 5. Vasculitis (Cutaneous Small Vessel Vasculitis / Leukocytoclastic Vasculitis)

Inflammation and destruction of blood vessel walls, manifesting as palpable purpura.

SUBSECTION: Clinical Features

• Non-blanching palpable purpura** — the hallmark lesion (purpuric papules and plaques that are raised and can be felt)

• Painful** (not pruritic) — distinguishes from many other rashes

• Predilection for **dependent areas**: Lower legs, ankles, buttocks (gravity-dependent distribution)

• May progress to **hemorrhagic bullae, necrosis, and ulceration** in severe cases

• Systemic involvement**: Joint pain, fever, renal (hematuria/proteinuria), GI (abdominal pain, bleeding)

• Diascopy**: **Non-blanching** — this is the key distinguishing feature from toxic erythema

Chapter 16

SUBSECTION: Etiology

SUBSECTION: Etiology

TABLE: Causes of Cutaneous Vasculitis

Category	Specific Causes
Infections	Hepatitis B, Hepatitis C, HIV, Streptococcal infection, Endocarditis
Autoimmune	SLE, Rheumatoid Arthritis, Sjögren's syndrome
Malignancy	Lymphoma, Leukemia, Myelodysplastic syndrome
Drugs	Antibiotics (penicillins, sulfonamides), NSAIDs, Thiazide diuretics, Allopurinol
Cryoglobulinemia	Often associated with Hepatitis C
Idiopathic	Up to 50% of cases — Henoch-Schönlein purpura (IgA vasculitis) in children

KEY TEACHING POINTS

IMPORTANT

- **Non-blanching palpable purpura = Vasculitis until proven otherwise** — this is the single most important clinical sign.

- **Diascopy is the key bedside test**: Press a glass slide against the lesion. If it does NOT blanch → vasculitis. If it blanches → toxic erythema or other causes.

- **Vessel wall inflammation** leads to extravasation of RBCs into the dermis — this is why the lesions are non-blanching and palpable.

- **Always check urinalysis** — renal involvement (glomerulonephritis) is a serious complication.

- **Biopsy**: Leukocytoclastic vasculitis — neutrophilic infiltration of vessel walls with fibrinoid necrosis and nuclear dust (leukocytoclasis).

Chapter 17

TABLE: Diascopy — The Critical Bedside Test

TABLE: Diascopy — The Critical Bedside Test

| Test | Blanching (Positive) | Non-Blanching (Negative) | |---|---|---| | **Interpretation** | Blood is in vessels (erythema) | Blood has extravasated into tissue (purpura) | | **Causes** | Toxic erythema, urticaria, viral exanthem | Vasculitis, meningococcemia, DIC, embolic phenomena | | **Clinical significance** | Benign, self-limiting | Potentially serious — investigate for systemic disease |

MCQ TRAPS & PITFALLS

- Do NOT call palpable purpura "allergic"** — it is vasculitis, not an allergic reaction. The term "hypersensitivity vasculitis" is outdated.
- Vasculitis is NOT always idiopathic** — always investigate for underlying infection (Hepatitis B/C), autoimmune disease, and malignancy.
- Henoch-Schönlein purpura (IgA vasculitis)** is the most common vasculitis in children — presents with palpable purpura, arthritis, abdominal pain, and nephritis.

OSCE PEARLS

- Always perform diascopy** — press a glass slide against the lesion. Non-blanching = vasculitis.
- Check urine dipstick** — hematuria and proteinuria indicate renal vasculitis.
- Ask about joint pain, abdominal pain, and recent infections** — these suggest systemic vasculitis.
- Biopsy within 24–48 hours** of lesion appearance — older lesions may show secondary changes that obscure the diagnosis.

Chapter 18

QUESTION:

QUESTION:

Q: A 55-year-old man presents with a 1-week history of painful, raised, purplish lesions on his lower legs and ankles. The lesions do not blanch when pressed with a glass slide. He also reports joint pain and dark-colored urine. Urinalysis shows hematuria and proteinuria. Serum creatinine is elevated at 180 μmol/L. What is the most likely diagnosis? A) Erythema nodosum B) Leukocytoclastic vasculitis with renal involvement C) Cellulitis D) Allergic contact dermatitis E) Psoriatic arthropathy ANSWER: B EXPLANATION: This presentation is classic for **leukocytoclastic vasculitis with renal involvement**: (1) **non-blanching palpable purpura** on dependent areas (lower legs/ankles) — confirmed by diascopy; (2) **painful** lesions (not pruritic); (3) **joint pain** (arthralgia); (4) **renal involvement** — hematuria, proteinuria, and elevated creatinine indicating glomerulonephritis. Erythema nodosum (A) presents as tender subcutaneous nodules on shins, not purpuric lesions. Cellulitis (C) is typically unilateral, warm, and erythematous — not purpuric. Allergic contact dermatitis (D) is pruritic and blanching. Psoriatic arthropathy (E) presents with joint pain and psoriatic plaques, not purpura. Investigations should include hepatitis B/C serology, ANA, ANCA, complement levels, and skin biopsy. Management: Treat underlying cause, consider systemic corticosteroids for severe or progressive disease. ---

CHAPTER: 3 — Skin Manifestations of Malignancy

SUBTITLE: Paraneoplastic Dermatoses and Cutaneous Metastases

Chapter 19

SECTION: Overview: Skin Signs of Internal Malignancy

SECTION: Overview: Skin Signs of Internal Malignancy

The skin can provide early clues to underlying malignancy. Recognizing paraneoplastic dermatoses can lead to earlier diagnosis and improved outcomes. These cutaneous manifestations are NOT caused by direct tumor invasion but by immune-mediated or hormone-mediated mechanisms.

KEY TEACHING POINTS

• Paraneoplastic dermatoses** are skin conditions associated with internal malignancy but NOT caused by direct tumor invasion.

• Timing**: The skin manifestation may precede, coincide with, or follow the diagnosis of malignancy.

• Screening**: When a paraneoplastic dermatosis is suspected, age-appropriate cancer screening is mandatory.

• Biopsy is essential** — many paraneoplastic dermatoses have characteristic histopathological features.

SECTION: 1. Acanthosis Nigricans (Malignant Type)

Chapter 20

SUBSECTION: Clinical Features

SUBSECTION: Clinical Features

• Velvety, hyperpigmented, thickened skin** — typically in flexural areas (axillae, groin, neck)

• Malignant acanthosis nigricans**: Rapid onset, extensive, may involve mucous membranes (lips, oral mucosa)

• Sign of Leser-Trélat**: Sudden eruption of multiple seborrheic keratoses — associated with internal malignancy

• Associated malignancy**: **Gastric adenocarcinoma** (most common), other GI malignancies

SUBSECTION: Benign vs. Malignant Acanthosis Nigricans

Feature	Benign AN	Malignant AN
Onset	Gradual (years)	Rapid (weeks to months)
Distribution	Flexures only	Flexures + mucous membranes + palms (tripe palms)
Age	Young, obese, diabetic	Older adults (>40 years)
Associated condition	Insulin resistance, obesity, PCOS, type 2 diabetes	Internal malignancy (gastric adenocarcinoma #1)
Management	Treat underlying metabolic condition	Urgent cancer screening

KEY TEACHING POINTS

• Malignant acanthosis nigricans is a paraneoplastic sign** — it is driven by tumor-derived growth factors (TGF-α, IGF-1) stimulating keratinocyte and fibroblast proliferation.

• Gastric adenocarcinoma is the most common associated malignancy** — always perform upper GI endoscopy.

• Sign of Leser-Trélat** (eruptive seborrheic keratoses) is a related paraneoplastic sign — think malignancy.

• Benign AN is far more common** — associated with insulin resistance, obesity, and type 2 diabetes. Do not over-investigate every case of AN.

Chapter 21

MCQ TRAPS & PITFALLS

MCQ TRAPS & PITFALLS

• Do NOT assume all acanthosis nigricans is benign** — rapid onset in an older adult should trigger cancer screening.

• Malignant AN is NOT caused by the tumor directly invading the skin** — it is a paraneoplastic phenomenon mediated by growth factors.

• Sign of Leser-Trélat is NOT the same as simple seborrheic keratoses** — the key is the sudden eruption of multiple lesions.

SECTION: 2. Dermatomyositis (DM)

An idiopathic inflammatory myopathy with characteristic skin manifestations.

SUBSECTION: Clinical Features

• Heliotrope rash**: Violaceous/purplish discoloration of the eyelids with periorbital edema

• Gottron's papules**: Violaceous, flat-topped papules over the knuckles (MCP, PIP joints)

• Gottron's sign**: Violaceous erythema over knuckles, elbows, knees

• Shawl sign**: Erythema over the upper back and shoulders (shawl distribution)

• V-sign**: Erythema over the anterior chest in a V-shape

• Mechanic's hands**: Cracked, fissured skin on the lateral aspects of fingers

• Proximal muscle weakness**: Difficulty rising from a chair, climbing stairs, lifting arms

• Calcinosis cutis**: Calcium deposits in the skin (more common in juvenile DM)

Chapter 22

SUBSECTION: Malignancy Association

SUBSECTION: Malignancy Association

• Adult dermatomyositis (age >40)** has a significant association with underlying malignancy

• Associated malignancies**: Ovarian cancer (#1 in women), Lung cancer, Gastric cancer, Lymphoma, Colorectal cancer

• Risk is highest in the first 3 years** after DM diagnosis

• Screening**: Age-appropriate cancer screening + CT chest/abdomen/pelvis, tumor markers (CA-125 for ovarian cancer)

KEY TEACHING POINTS

• Dermatomyositis in adults >40 years = Cancer screening mandatory** — the association is well-established.

• Ovarian cancer is the most common associated malignancy in women** — always check CA-125 and pelvic imaging.

• Gottron's papules and heliotrope rash are pathognomonic** — if you see these, think DM.

• Juvenile DM is NOT associated with malignancy** — but has a higher risk of calcinosis and vasculitis.

• Anti-TIF1-γ (transcription intermediary factor 1-gamma) antibodies** are strongly associated with malignancy in DM — consider testing.

TABLE: Dermatomyositis — Skin Findings and Associated Malignancies

Skin Finding	Description	Associated Malignancy
Heliotrope rash	Violaceous eyelid discoloration	Ovarian, Lung, GI
Gottron's papules	Violaceous papules over knuckles	Ovarian, Lung, GI
Shawl sign	Upper back/shoulder erythema	Ovarian, Lung, GI
Mechanic's hands	Fissured lateral finger skin	Anti-synthetase syndrome (not malignancy-specific)
Calcinosis cutis	Skin calcification	Juvenile DM (not malignancy)

Chapter 23

MCQ TRAPS & PITFALLS

MCQ TRAPS & PITFALLS

• Do NOT confuse DM with polymyositis (PM)** — PM has NO skin findings. DM has both skin and muscle involvement.

• DM can occur WITHOUT muscle weakness** — this is called "amyopathic dermatomyositis" or "dermatomyositis sine myositis" — still carries malignancy risk.

• Heliotrope rash is NOT the same as periorbital allergic dermatitis** — the violaceous color and associated muscle weakness distinguish it.

OSCE PEARLS

OSCE PEARLS

- **Always examine the hands** — Gottron's papules over the knuckles are pathognomonic.

- **Check for proximal muscle weakness** — ask the patient to rise from a chair without using arms.

- **In an OSCE station, mention cancer screening** — this shows clinical awareness of the DM-malignancy association.

- **Order CK (creatine kinase) and LDH** — these are elevated in active myositis.

QUESTION:

IMPORTANT

Q: A 60-year-old woman presents with a 2-month history of progressive difficulty climbing stairs and rising from a chair. On examination, she has violaceous discoloration of both eyelids with periorbital edema and violaceous papules over her knuckles. CK is elevated at 2,500 U/L. What is the most important next step in management?

A) Start high-dose prednisolone

B) Perform skin biopsy

C) Screen for underlying malignancy

D) Order EMG and nerve conduction studies

E) Start methotrexate

ANSWER: C

EXPLANATION: This presentation is classic for **dermatomyositis**: (1) **heliotrope rash** (violaceous eyelid discoloration with periorbital edema); (2) **Gottron's papules** (violaceous papules over knuckles); (3) **proximal muscle weakness** (difficulty climbing stairs, rising from chair); (4) **elevated CK** (2,500 U/L — indicating active myositis). In a **60-year-old woman**, the most critical next step is to **screen for underlying malignancy** (C) — particularly **ovarian cancer** (the most common associated malignancy in women). While starting prednisolone (A) and performing skin biopsy (B) are important, malignancy screening takes priority because early detection significantly impacts prognosis. Investigations should include CT chest/abdomen/pelvis, CA-125, and age-appropriate cancer screening. EMG (D) and methotrexate (E) are part of the workup and management but are not the most urgent next step.

Chapter 24

SECTION: 3. Necrolytic Migratory Erythema (NME)

SECTION: 3. Necrolytic Migratory Erythema (NME)

A characteristic rash associated with **glucagonoma** — a rare pancreatic neuroendocrine tumor.

SUBSECTION: Clinical Features

- Erythematous, erosive, migratory rash** — begins in the groin/perineum → spreads to extremities
- Glossitis, angular cheilitis, and stomatitis** — oral mucosal involvement
- Weight loss, diarrhea, and diabetes mellitus** — systemic features
- "Super-glue" appearance**: The rash has a characteristic shiny, crusted, eroded appearance
- Associated with glucagonoma syndrome**: NME + diabetes + weight loss + diarrhea + glossitis

KEY TEACHING POINTS

- NME is the cutaneous manifestation of glucagonoma** — a pancreatic alpha-cell tumor.
- Glucagonoma triad**: NME + diabetes mellitus + weight loss.
- The rash is caused by amino acid deficiency** — excess glucagon drives protein catabolism.
- Diagnosis**: Elevated serum glucagon (>500 pg/mL), CT/MRI pancreas for tumor localization.
- Treatment**: Somatostatin analogs (Octreotide) for symptom control, surgical resection if localized.

Chapter 25

MCQ TRAPS & PITFALLS

MCQ TRAPS & PITFALLS

• Do NOT confuse NME with pemphigus vulgaris** — NME is erosive but does NOT show acantholysis on biopsy.

• NME is NOT a fungal infection** — despite the erythematous, crusted appearance in the groin, KOH preparation is negative.

• Glucagonoma is NOT the only cause of NME** — it can also occur in liver disease, celiac disease, and other malabsorptive states (pseudoglucagonoma syndrome).

SECTION: 4. Sweet Syndrome (Acute Febrile Neutrophilic Dermatosis)

SUBSECTION: Clinical Features

• Tender, erythematous plaques and nodules** — often with a "juicy" or vesicular surface

• Fever and leukocytosis** — systemic features

• Predilection for face, neck, and upper extremities**

• Associated with**: Upper respiratory tract infections, IBD, hematological malignancy (especially AML), pregnancy

• Histopathology**: Dense neutrophilic infiltrate in the dermis WITHOUT vasculitis

Chapter 26

KEY TEACHING POINTS

KEY TEACHING POINTS

• Sweet syndrome is a reactive neutrophilic dermatosis** — NOT an infection.

• AML (Acute Myeloid Leukemia) is the most common malignancy association** — always check FBC with differential.

• Treatment**: Systemic corticosteroids (Prednisolone 1mg/kg/day) — rapid response is characteristic.

• Recurrent Sweet syndrome** → Think underlying malignancy → Investigate thoroughly.

CHAPTER: 4 — Skin Manifestations of Endocrine Disease

SUBTITLE: Cutaneous Clues to Hormonal Disorders

Chapter 27

SECTION: 1. Acanthosis Nigricans (Benign/Metabolic Type)

SECTION: 1. Acanthosis Nigricans (Benign/Metabolic Type)

As discussed in Chapter 3, benign AN is associated with insulin resistance, obesity, and type 2 diabetes. The key distinguishing features from malignant AN are gradual onset, flexural distribution only, and association with metabolic syndrome.

KEY TEACHING POINTS

• Insulin resistance is the primary driver** — insulin and IGF-1 stimulate keratinocyte proliferation.

• PCOS (Polycystic Ovary Syndrome)** is a common association in young women — consider screening for AN in PCOS patients.

• Treatment**: Weight loss, metformin, topical retinoids for cosmetic improvement.

SECTION: 2. Pretibial Myxedema (Thyroid Dermopathy)

Chapter 28

SUBSECTION: Clinical Features

SUBSECTION: Clinical Features

• Waxy, indurated, non-pitting edema** — typically on the **pretibial area** (shins)

• "Peau d'orange" appearance** — orange-peel texture with prominent hair follicles

• Associated with Graves' disease** (hyperthyroidism) — NOT hypothyroidism

• May also affect the feet and toes** (thyroid acropachy)

• Caused by glycosaminoglycan (GAG) deposition** in the dermis — stimulated by TSH receptor antibodies

KEY TEACHING POINTS

• Pretibial myxedema is specific to Graves' disease** — it is caused by the same TSH receptor antibodies that cause hyperthyroidism.

• It does NOT correlate with thyroid function** — patients may be euthyroid or even hypothyroid when the skin changes appear.

• Treatment**: Topical high-potency corticosteroids (Clobetasol propionate 0.05% cream), compression stockings. Most cases are mild and self-limiting.

MCQ TRAPS & PITFALLS

• Do NOT confuse pretibial myxedema with pitting edema of heart failure** — pretibial myxedema is NON-pitting and has a waxy, indurated texture.

• Pretibial myxedema is NOT caused by hypothyroidism** — it is associated with Graves' disease (hyperthyroidism).

Chapter 29

SECTION: 3. Addison's Disease — Hyperpigmentation

SECTION: 3. Addison's Disease — Hyperpigmentation

SUBSECTION: Clinical Features

• Generalized hyperpigmentation** — especially in sun-exposed areas, skin creases, buccal mucosa, scars, and pressure points

• Caused by elevated ACTH** — ACTH stimulates melanocytes via the melanocortin-1 receptor

• Associated symptoms**: Fatigue, weight loss, hypotension, salt craving, hypoglycemia

• Primary adrenal insufficiency** — autoimmune adrenalitis is the most common cause in developed countries

KEY TEACHING POINTS

• Hyperpigmentation in Addison's disease is due to ACTH excess** — the same precursor (POMC) produces both ACTH and MSH (melanocyte-stimulating hormone).

• Buccal mucosal pigmentation is a key clinical sign** — always examine the mouth in suspected Addison's.

• New or darkened scars** are a classic feature — ask about pigmentation in surgical scars.

• Diagnosis**: Short Synacthen test (ACTH stimulation test) — failure of cortisol to rise confirms adrenal insufficiency.

Chapter 30

CHAPTER: 5 — Skin Manifestations of Gastrointestinal Disease

CHAPTER: 5 — Skin Manifestations of Gastrointestinal Disease

SUBTITLE: Dermatological Clues to GI Disorders

SECTION: 1. Dermatitis Herpetiformis (DH)

The cutaneous manifestation of **coeliac disease** — an autoimmune gluten-sensitive enteropathy.

Chapter 31

SUBSECTION: Clinical Features

SUBSECTION: Clinical Features

• Intensely pruritic, grouped vesicles and papules** — on extensor surfaces (elbows, knees, buttocks, scalp)

• Symmetric distribution** — a hallmark feature

• Excoriations are more commonly seen than intact vesicles** — because the itch is so intense

• Associated with coeliac disease** — virtually all DH patients have gluten-sensitive enteropathy (though may be asymptomatic)

• IgA deposition in dermal papillae** on direct immunofluorescence — pathognomonic

KEY TEACHING POINTS

• DH is the skin manifestation of coeliac disease** — all DH patients should be screened for coeliac disease (anti-tTG IgA antibodies).

• The itch is EXCRUCIATING** — patients often present with excoriations rather than intact vesicles.

• Dapsone is the treatment of choice** — rapid relief of itch within 24–48 hours. If dapsone works, it confirms the diagnosis.

• Gluten-free diet** is the definitive long-term treatment — reduces both skin and intestinal disease.

• Associated with other autoimmune conditions**: Type 1 diabetes, thyroid disease, pernicious anemia.

MCQ TRAPS & PITFALLS

• Do NOT confuse DH with scabies** — both cause intense itch, but DH has a characteristic distribution on extensor surfaces and is associated with coeliac disease.

• DH vesicles are NOT caused by herpes virus** — the name is misleading. It is an IgA-mediated autoimmune condition.

• Dapsone response is diagnostic** — if the itch resolves within 48 hours of starting dapsone, DH is confirmed.

Chapter 32

SECTION: 2. Aphthous Stomatitis and GI Disease

SECTION: 2. Aphthous Stomatitis and GI Disease

Recurrent aphthous ulcers are associated with several GI conditions: • **Coeliac disease** — aphthous ulcers may be the only presenting feature • **IBD (Crohn's disease, Ulcerative Colitis)** — oral ulcers may parallel disease activity • **Behçet's disease** — recurrent oral ulcers are a diagnostic criterion ---

CHAPTER: 6 — Summary and High-Yield Review

SUBTITLE: Essential Facts for Clinical Examinations

Chapter 33

SECTION: Master Table — Skin Lesions and Systemic Disease

SECTION: Master Table — Skin Lesions and Systemic Disease

TABLE: Comprehensive Summary — Cutaneous Manifestations of Systemic Disease

Skin Lesion	Morphology	Key Associations	Most Common Cause	Key Investigation
Erythema multiforme	Target lesions (3 zones)	HSV, Mycoplasma	HSV	Clinical diagnosis
Pyoderma gangrenosum	Painful ulcer, undermined purple borders	IBD, RA, malignancy	IBD	Biopsy (exclusion)
Erythema nodosum	Tender shin nodules	Strep, TB, Sarcoidosis, IBD	Streptococcal infection	ASO titer, CXR
Vasculitis	Non-blanching palpable purpura	Hep B/C, SLE, lymphoma	Context-dependent	Biopsy, urinalysis
Acanthosis nigricans	Velvety hyperpigmentation	Insulin resistance, gastric cancer	Metabolic syndrome	Glucose, endoscopy
Dermatomyositis	Heliotrope rash, Gottron's papules	Ovarian, lung, GI cancer	Autoimmune	CK, malignancy screen
Necrolytic migratory erythema	Erosive migratory rash	Glucagonoma	Pancreatic tumor	Serum glucagon
Sweet syndrome	Tender erythematous plaques	AML, IBD, infection	Reactive	FBC, biopsy
Dermatitis herpetiformis	Pruritic vesicles (extensor)	Coeliac disease	Gluten sensitivity	Anti-tTG IgA, biopsy
Pretibial myxedema	Waxy non-pitting edema	Graves' disease	Autoimmune thyroid	TSH, TSH-R antibodies

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SECTION: The Diascopy Test — A Critical Clinical Skill

Chapter 34

KEY TEACHING POINTS

KEY TEACHING POINTS

IMPORTANT

- **Diascopy** (glass press test) is the single most important bedside test for differentiating blanching from non-blanching rashes.

- **Technique**: Press a clear glass slide firmly against the skin lesion.

- **Blanching (positive)**: Lesion disappears → blood is within vessels → toxic erythema, urticaria, viral exanthem.

- **Non-blanching (negative)**: Lesion persists → blood has extravasated into tissue → vasculitis, purpura, embolic phenomena.

- **Clinical significance**: Non-blanching palpable purpura = vasculitis until proven otherwise → investigate for systemic disease.

SECTION: Approach to a Patient with Rash and Systemic Symptoms

KEY TEACHING POINTS

When evaluating a patient with rash AND systemic symptoms, follow this systematic approach: 1. **Characterize the rash**: Morphology, distribution, blanching vs. non-blanching, painful vs. pruritic 2. **Perform diascopy**: Glass press test to differentiate blanching from non-blanching 3. **Assess for systemic involvement**: Joint pain, fever, weight loss, renal (urinalysis), GI symptoms 4. **Targeted investigations based on clinical suspicion**: - Suspected vasculitis → Hepatitis B/C serology, ANA, ANCA, complement, biopsy - Suspected EN → ASO titer, CXR, stool cultures - Suspected DM → CK, malignancy screening, anti-TIF1-γ - Suspected PG → Biopsy (rule out infection), colonoscopy if GI symptoms 5. **Biopsy when diagnosis is uncertain** — but remember that many conditions are clinical diagnoses ---

Chapter 35

QUESTION:

QUESTION:

Q: A 50-year-old man presents with a 2-week history of a widespread, symmetrical, blanching maculopapular rash. He was diagnosed with infectious mononucleosis 10 days ago and was started by his GP on amoxicillin. He has no systemic symptoms other than mild pruritus. What is the most likely cause of his rash? A) Anaphylaxis to amoxicillin B) Toxic erythema secondary to EBV-amoxicillin interaction C) Measles D) Scarlet fever E) Acute HIV seroconversion ANSWER: B EXPLANATION: This is a classic presentation of **toxic erythema secondary to EBV-amoxicillin interaction**. The key features are: (1) **EBV infection (infectious mononucleosis)** treated with **amoxicillin** — up to 90% of EBV patients given penicillin-class antibiotics develop a rash; (2) **widespread symmetrical maculopapular rash** — the classic morbilliform pattern of toxic erythema; (3) **blanching** on diascopy — distinguishing it from vasculitis; (4) **mild pruritus without systemic toxicity** — consistent with a benign, self-limiting drug eruption. This is NOT anaphylaxis (A) — there is no urticaria, angioedema, or respiratory compromise. Measles (C) presents with cough, coryza, conjunctivitis, and Koplik spots. Scarlet fever (D) has a sandpaper-like rash with desquamation. Acute HIV seroconversion (E) can cause a similar rash but the EBV-amoxicillin association is far more likely in this context. Management: Supportive — the rash will resolve spontaneously. Avoid penicillin-class antibiotics in future EBV infections. ---

QUESTION:

Q: A 7-year-old boy presents with palpable purpura on his lower legs and buttocks, abdominal pain, and joint swelling. Urinalysis shows microscopic hematuria. He had an upper respiratory tract infection 2 weeks ago. What is the most likely diagnosis? A) Leukocytoclastic vasculitis secondary to SLE B) Henoch-Schönlein purpura (IgA vasculitis) C) Meningococcemia D) Idiopathic thrombocytopenic purpura E) Erythema nodosum ANSWER: B EXPLANATION: This is a classic presentation of **Henoch-Schönlein purpura (HSP) / IgA vasculitis** — the most common vasculitis in children. The classic tetrad is: (1) **palpable purpura** on dependent areas (lower legs, buttocks); (2) **arthritis/arthralgia** (joint swelling); (3) **abdominal pain** (GI vasculitis); (4) **renal involvement** (hematuria). The preceding upper respiratory tract infection is a common trigger. HSP is an **IgA-mediated small vessel vasculitis**. SLE (A) is extremely rare in a 7-year-old boy and would have additional features (malar rash, positive ANA). Meningococcemia (C) causes rapidly progressive purpura with septic shock — this child is not acutely unwell. ITP (D) causes thrombocytopenia and petechiae, not palpable purpura with systemic features. Erythema nodosum (E) presents as tender subcutaneous nodules, not purpura. Management of HSP is supportive — most cases resolve spontaneously. Monitor renal function for nephritis. ---

QUESTION:

Q: A 45-year-old woman presents with intensely pruritic, grouped vesicles on her elbows, knees, and buttocks. She has been scratching so much that most lesions are now excoriations. She reports intermittent diarrhea. Anti-tissue transglutaminase (anti-tTG) IgA antibodies are positive. What is the most appropriate treatment? A) Oral acyclovir B) Oral dapsone and gluten-free diet C) Topical corticosteroids only D) Oral fluconazole E) Systemic corticosteroids ANSWER: B EXPLANATION: This presentation is classic for **dermatitis herpetiformis (DH)**: (1) **intensely pruritic grouped vesicles** on extensor surfaces (elbows, knees, buttocks); (2) **excoriations more common than intact vesicles** due to intense itch; (3) **associated with coeliac disease** — positive anti-tTG IgA antibodies and diarrhea; (4) **symmetric distribution**. The treatment of choice is **oral dapsone** (B) — it provides rapid relief of itch within 24–48 hours. A **gluten-free diet** is the definitive long-term treatment. Acyclovir (A) is incorrect — DH is NOT caused by herpes virus despite the name. Topical corticosteroids (C) are insufficient for DH. Fluconazole (D) is for fungal infections. Systemic corticosteroids (E) are not first-line for DH. The rapid response to dapsone is virtually diagnostic of DH. ---

Chapter 36

QUESTION:

QUESTION:

Q: A 55-year-old man presents with a 3-month history of velvety, hyperpigmented skin in his axillae and neck. The changes have been gradually worsening. He is obese (BMI 34) and has a family history of type 2 diabetes. Fasting glucose is 8.5 mmol/L (normal <6.1). HbA1c is 7.2%. What is the most likely diagnosis? A) Malignant acanthosis nigricans B) Benign acanthosis nigricans C) Addison's disease D) Haemochromatosis E) Cushing's syndrome ANSWER: B EXPLANATION: This presentation is classic for **benign acanthosis nigricans**: (1) **gradual onset** over 3 months — consistent with benign AN (malignant AN has rapid onset); (2) **flexural distribution** (axillae, neck) — without mucosal involvement; (3) **associated with obesity** (BMI 34) and **insulin resistance** (elevated fasting glucose 8.5 mmol/L, HbA1c 7.2% — indicating type 2 diabetes). Malignant AN (A) would have rapid onset, mucosal involvement, and occur in older adults without metabolic risk factors. Addison's disease (C) causes generalized hyperpigmentation, especially in buccal mucosa and scars — not velvety flexural changes. Haemochromatosis (D) causes bronze pigmentation. Cushing's syndrome (E) causes striae, central obesity, and thin skin — not velvety hyperpigmentation. Management: Weight loss, metformin for diabetes, and topical retinoids for cosmetic improvement. ---

CHAPTER: 7 — Quick Reference Tables

SUBTITLE: High-Yield Comparisons for Exam Preparation

Chapter 37

TABLE: Blanching vs. Non-Blanching Rashes

TABLE: Blanching vs. Non-Blanching Rashes

Feature	Blanching Rash	Non-Blanching Rash
Diascopy	Lesion disappears	Lesion persists
Pathophysiology	Vasodilation (blood in vessels)	Extravasation of RBCs (blood in tissue)
Examples	Toxic erythema, urticaria, viral exanthem, scarlet fever	Vasculitis, purpura, meningococcemia, DIC
Clinical significance	Usually benign	Potentially serious — investigate
Key test	Glass press test	Glass press test

TABLE: Painful vs. Pruritic Skin Lesions

Feature	Painful Lesions	Pruritic Lesions
Examples	Vasculitis, pyoderma gangrenosum, erythema nodosum, cellulitis	Dermatitis herpetiformis, urticaria, allergic contact dermatitis, scabies
Key associations	Vasculitis, infection, neutrophilic dermatoses	Allergic, autoimmune (coeliac), parasitic
Clinical approach	Investigate for systemic disease, infection	Consider allergy, autoimmune, infestation

TABLE: Skin Lesions by Distribution

Distribution	Skin Lesions
Acral (hands, feet)	Erythema multiforme, chilblains, vasculitis
Extensor surfaces (elbows, knees)	Dermatitis herpetiformis, psoriasis, Gottron's papules
Flexural (axillae, groin)	Acanthosis nigricans, intertrigo, candidiasis
Pretibial (shins)	Erythema nodosum, pretibial myxedema, necrobiosis lipoidica
Dependent areas (legs, buttocks)	Vasculitis, dependent edema
Face and neck	Dermatomyositis (heliotrope), SLE (malar rash), rosacea

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Chapter 38

CHAPTER: 8 — Final Examination Preparation

CHAPTER: 8 — Final Examination Preparation

SUBTITLE: Last-Minute Review and Key Facts

SECTION: Top 20 High-Yield Facts

Chapter 39

KEY TEACHING POINTS

KEY TEACHING POINTS

IMPORTANT

1. **Non-blanching palpable purpura = Vasculitis** until proven otherwise

2. **EBV + ampicillin → rash in up to 90%** of patients

3. **Target lesions = Erythema multiforme** — most common cause is HSV

4. **Pyoderma gangrenosum shows pathergy** — do NOT debride

5. **Erythema nodosum is the most common panniculitis** — most common cause is streptococcal infection

6. **Dermatitis herpetiformis = Coeliac disease** — treat with dapsone and gluten-free diet

7. **Dermatomyositis in adults >40 = Cancer screening mandatory** — ovarian cancer #1 in women

8. **Necrolytic migratory erythema = Glucagonoma** — check serum glucagon

9. **Sweet syndrome = AML association** — check FBC

10. **Pretibial myxedema = Graves' disease** — NOT hypothyroidism

11. **Acanthosis nigricans: rapid onset + older adult = Malignancy** — gastric adenocarcinoma #1

12. **Addison's hyperpigmentation = ACTH excess** — check buccal mucosa

13. **EM is Type IV hypersensitivity** — NOT Type I

14. **Diascopy (glass press test)** is the most important bedside test for rash evaluation

15. **Heliotrope rash + Gottron's papules = Dermatomyositis**

16. **Sign of Leser-Trélat (eruptive seborrheic keratoses) = Internal malignancy**

17. **Henoch-Schönlein purpura = Most common vasculitis in children**

18. **PG is a diagnosis of exclusion** — always rule out infection and malignancy

19. **EN does NOT ulcerate** — if ulceration present, consider alternative diagnosis

20. **Toxic erythema is self-limiting** — management is supportive

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SECTION: Common MCQ Themes

MCQ TRAPS & PITFALLS

• Theme 1: Diascopy** — Questions will test your ability to differentiate blanching from non-blanching rashes. Always think: "Does it blanch?"

• Theme 2: Drug-rash associations** — EBV + ampicillin, sulfonamides + SJS/TEN, allopurinol + DRESS syndrome

• Theme 3: Malignancy screening** — DM, AN, Sweet syndrome, NME all have malignancy associations

• Theme 4: Pathergy** — PG worsens with debridement. Do NOT operate on these ulcers.

• Theme 5: Hypersensitivity types** — EM is Type IV, NOT Type I. Urticaria is Type I.

• Theme 6: Distribution patterns** — Acral (EM), extensor (DH), flexural (AN), pretibial (EN, myxedema), dependent (vasculitis)

Chapter 40

SECTION: OSCE Station Preparation

SECTION: OSCE Station Preparation

OSCE PEARLS

OSCE PEARLS

- **When examining a rash, always perform diascopy** — this is expected in OSCE stations and demonstrates clinical competence.

- **Describe lesions using correct terminology**: macule, papule, plaque, nodule, vesicle, bulla, pustule, ulcer, target lesion, purpura.

- **Always examine the mouth** — mucosal involvement changes the diagnosis (EM major vs. minor, Addison's, DH).

- **Ask about systemic symptoms** — joint pain, fever, weight loss, GI symptoms, urinary symptoms.

- **Mention malignancy screening** when presenting DM, AN, or Sweet syndrome — this shows clinical awareness.

[redacted]

End of Document

This comprehensive clinical summary covers the essential knowledge required for recognizing cutaneous manifestations of systemic disease. Mastery of these concepts, combined with the ability to perform diascopy and recognize pathognomonic skin lesions, will equip you for both written examinations and clinical practice.